In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
```

In [2]:
```
diabetes_df = pd.read_csv("diabetes.csv")

print(diabetes_df.head())
```

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
```

In [3]:
```
diabetes_df
```

Out[3]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Pregnancies</th> <th>Glucose</th> <th>BloodPressure</th> <th>SkinThickness</th> <th>Insulin</th> <th>BMI</th> <th>DiabetesPedigreeFunction</th> <th>Age</th> <th>Outcome</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>...</th> <th>763</th> <th>764</th> <th>765</th> <th>766</th> <th>767</th> </tbody> <tbody></tbody>

6	148	72	35	0	33.6	0.627	50	1
1	85	66	29	0	26.6	0.351	31	0
8	183	64	0	0	23.3	0.672	32	1
1	89	66	23	94	28.1	0.167	21	0
0	137	40	35	168	43.1	2.288	33	1
...	...	...	...	...	...	...	...	...
10	101	76	48	180	32.9	0.171	63	0
2	122	70	27	0	36.8	0.340	27	0
5	121	72	23	112	26.2	0.245	30	0
1	126	60	0	0	30.1	0.349	47	1
1	93	70	31	0	30.4	0.315	23	0

768 rows × 9 columns

In [4]:
```
diabetes_df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Pregnancies               768 non-null    int64  
 1   Glucose                   768 non-null    int64  
 2   BloodPressure             768 non-null    int64  
 3   SkinThickness             768 non-null    int64  
 4   Insulin                   768 non-null    int64  
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64  
 8   Outcome                   768 non-null    int64  
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

In [5]:
```
diabetes_df.describe()
```

Out[5]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Pregnancies</th> <th>Glucose</th> <th>BloodPressure</th> <th>SkinThickness</th> <th>Insulin</th> <th>BMI</th> <th>DiabetesPedigreeFunction</th> <th>Age</th> <th>Outcome</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75%</th> <th>max</th> </tbody> <tbody></tbody>

768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	
3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	0.471876	
3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	0.331329	
0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	
1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.243750	
3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	0.372500	
6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	
17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	2.420000	

In [6]:
```
print("Dataset Shape:", diabetes_df.shape)
```

```
Dataset Shape: (768, 9)
```

In [7]:
```
print(diabetes_df.isnull().sum())
```

```
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64
```

In [8]:
```
print("Column Names:")
print(diabetes_df.columns)
```

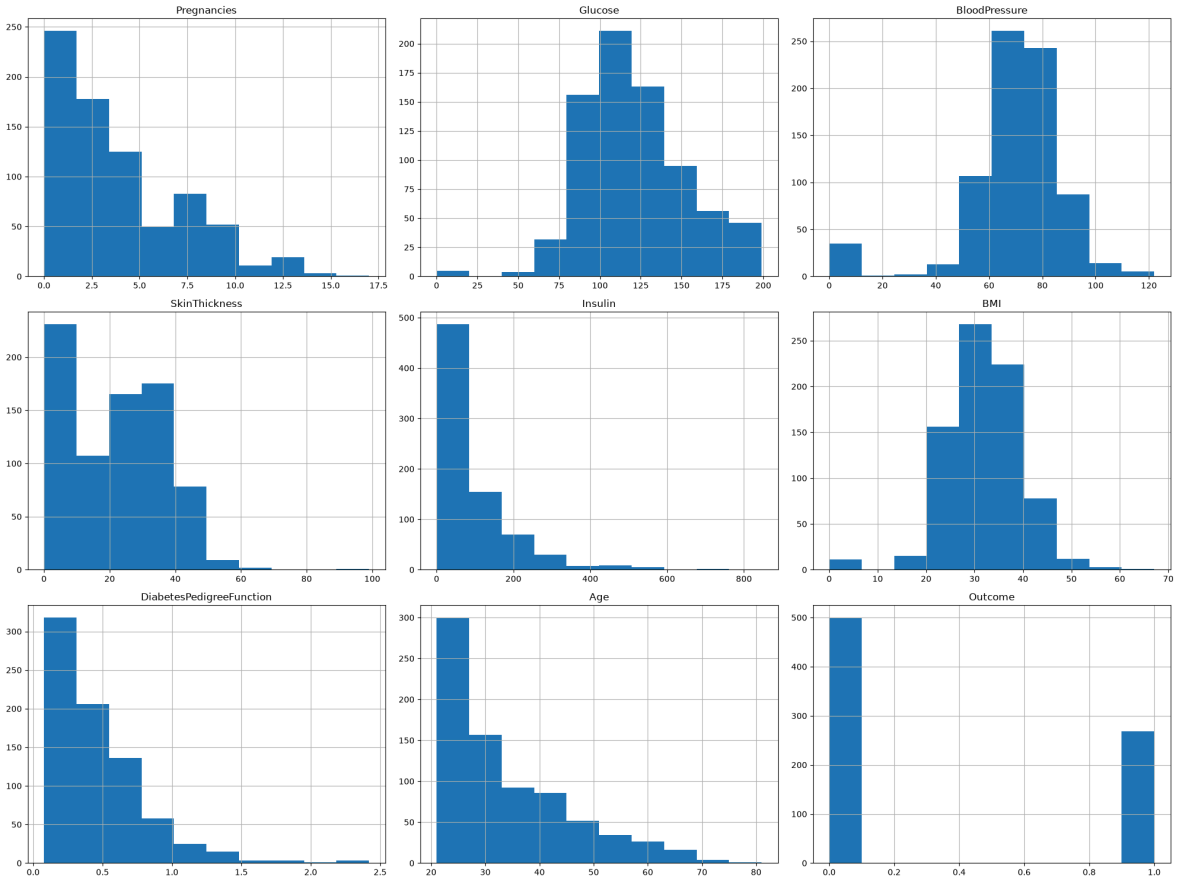
```
Column Names:
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insuli
n',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='str')
```

In [9]:
```
print(diabetes_df["Outcome"].value_counts())
```

```
Outcome
0    500
1    268
Name: count, dtype: int64
```

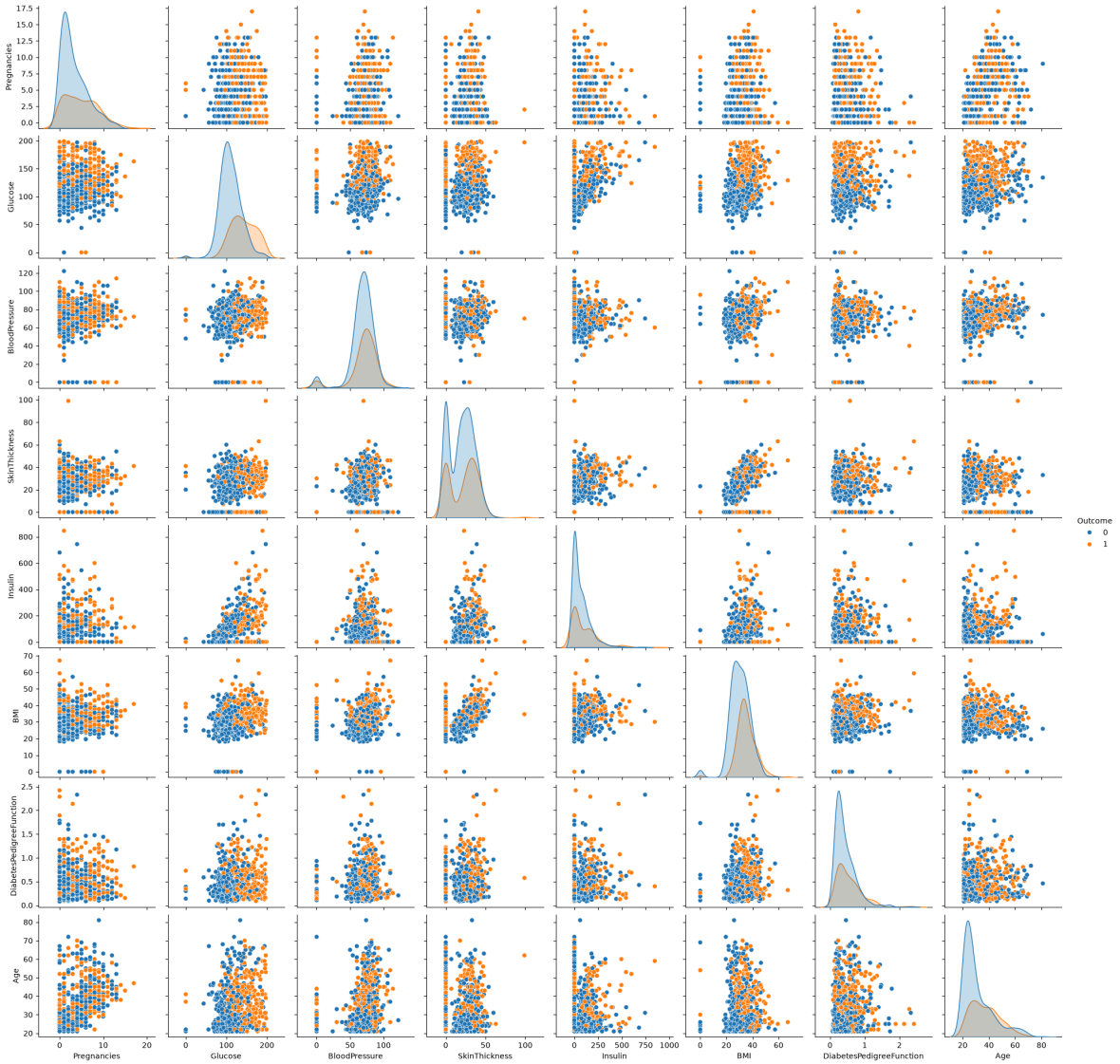
In [10]:
```
diabetes_df.hist(figsize=(20, 15))
```

```
plt.tight_layout()
plt.show()
```



In [11]:
```
sns.pairplot(diabetes_df, hue="Outcome")

plt.show()
```



In [12]:
```
correlation_matrix = diabetes_df.corr(numeric_only=True)

print(correlation_matrix)
```

```
                          Pregnancies   Glucose  BloodPressure  SkinThickn
ess  \
Pregnancies                  1.000000  0.129459       0.141282      -0.081
672
Glucose                      0.129459  1.000000       0.152590       0.057
328
BloodPressure                0.141282  0.152590       1.000000       0.207
371
SkinThickness               -0.081672  0.057328       0.207371       1.000
000
Insulin                     -0.073535  0.331357       0.088933       0.436
783
BMI                          0.017683  0.221071       0.281805       0.392
573
DiabetesPedigreeFunction    -0.033523  0.137337       0.041265       0.183
928
Age                          0.544341  0.263514       0.239528      -0.113
970
Outcome                      0.221898  0.466581       0.065068       0.074
752

                           Insulin       BMI  DiabetesPedigreeFunction  \
Pregnancies              -0.073535  0.017683                 -0.033523
Glucose                   0.331357  0.221071                  0.137337
BloodPressure             0.088933  0.281805                  0.041265
SkinThickness             0.436783  0.392573                  0.183928
Insulin                   1.000000  0.197859                  0.185071
BMI                       0.197859  1.000000                  0.140647
DiabetesPedigreeFunction  0.185071  0.140647                  1.000000
Age                      -0.042163  0.036242                  0.033561
Outcome                   0.130548  0.292695                  0.173844

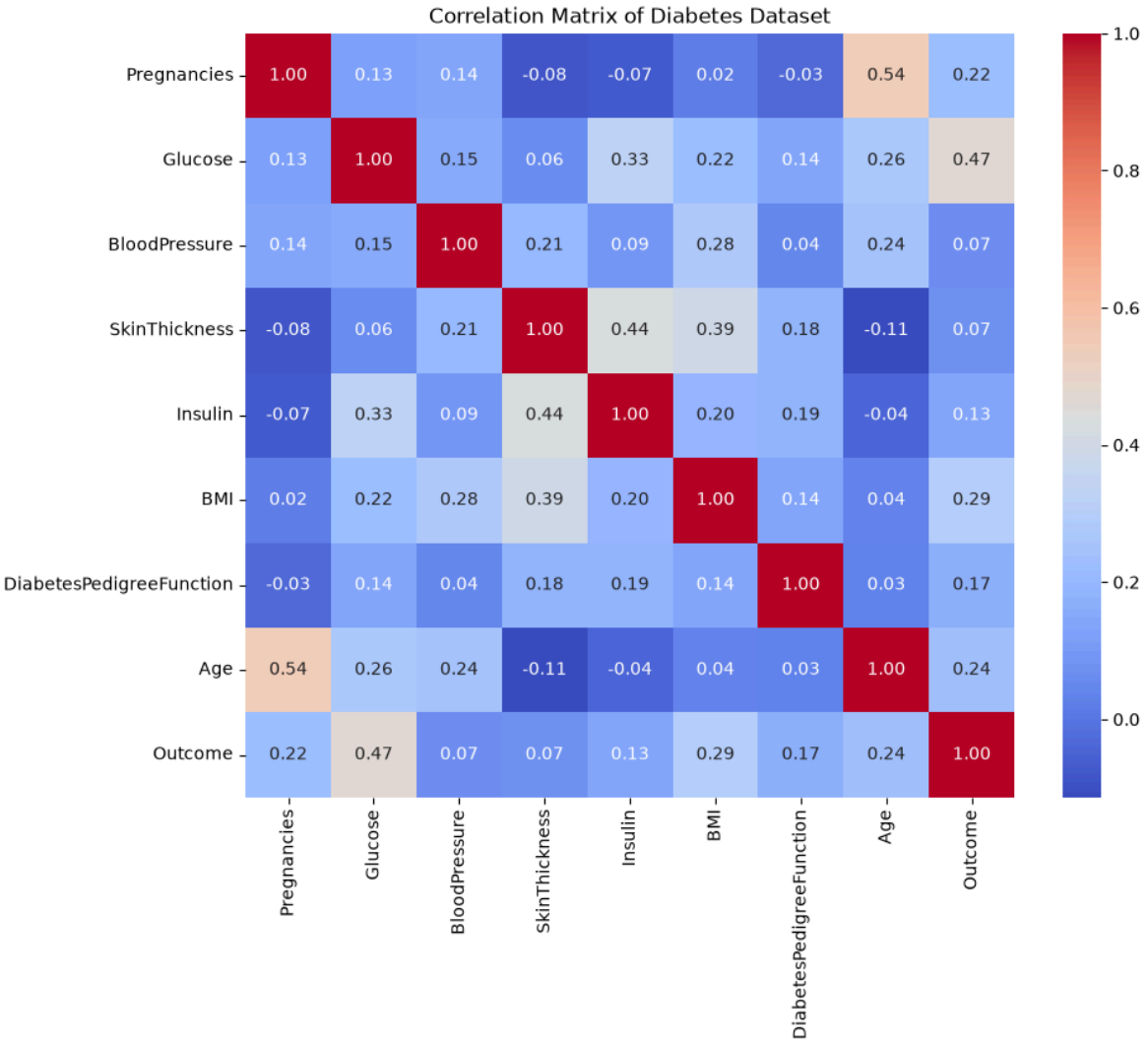
                               Age   Outcome
Pregnancies               0.544341  0.221898
Glucose                   0.263514  0.466581
BloodPressure             0.239528  0.065068
SkinThickness            -0.113970  0.074752
Insulin                  -0.042163  0.130548
BMI                       0.036242  0.292695
DiabetesPedigreeFunction  0.033561  0.173844
Age                       1.000000  0.238356
Outcome                   0.238356  1.000000
```

In [13]:

```
plt.figure(figsize=(10, 8))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Matrix of Diabetes Dataset")
plt.show()
```



In [14]:
```
print(diabetes_df.head(10))
```

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1
5            5      116             74              0        0  25.6
6            3       78             50             32       88  31.0
7           10      115              0              0        0  35.3
8            2      197             70             45      543  30.5
9            8      125             96              0        0   0.0

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
5                     0.201   30        0
6                     0.248   26        1
7                     0.134   29        0
8                     0.158   53        1
9                     0.232   54        1
```

In [ ]: